

# Optimization Homework 3

Dai Xunlian 225040015

225040015@link.cuhk.edu.cn

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## 1 Problem 1

### 1.1 (a) Dual Formulations

#### 1.1.1 First Linear Program

**Primal (Minimization):**

$$\min \quad 5x_1 + x_2 - 4x_3 \quad (1)$$

$$\text{s.t.} \quad x_1 + x_2 + x_3 + x_4 \geq 19 \quad (2)$$

$$4x_2 + 8x_4 \leq 55 \quad (3)$$

$$x_1 + 6x_2 - x_3 = 7 \quad (4)$$

$$x_1 \text{ free, } x_2 \geq 0, x_3 \geq 0, x_4 \leq 0 \quad (5)$$

**Dual (Maximization):** Let  $y_1, y_2, y_3$  denote the dual variables corresponding to the three primal constraints respectively.

$$\max \quad 19y_1 + 55y_2 + 7y_3 \quad (6)$$

$$\text{s.t.} \quad y_1 + y_3 = 5 \quad (7)$$

$$y_1 + 4y_2 + 6y_3 \leq 1 \quad (8)$$

$$y_1 - y_3 \leq -4 \quad (9)$$

$$y_1 + 8y_2 \geq 0 \quad (10)$$

$$y_1 \geq 0, y_2 \leq 0, y_3 \text{ free} \quad (11)$$

### 1.1.2 Second Linear Program

**Primal (Maximization):**

$$\max \quad 2x_1 - 7x_2 + 6x_3 + 5x_4 \quad (12)$$

$$\text{s.t.} \quad 2x_1 - 3x_2 - 5x_3 - 4x_4 \leq 20 \quad (13)$$

$$7x_1 + 2x_2 + 6x_3 - 2x_4 = 35 \quad (14)$$

$$4x_1 + 5x_2 - 3x_3 - 2x_4 \geq 15 \quad (15)$$

$$x_1 \leq 10 \quad (16)$$

$$x_2 \leq 5 \quad (17)$$

$$x_3 \geq 2 \quad (18)$$

$$x_1, x_2, x_3, x_4 \geq 0 \quad (19)$$

**Dual (Minimization):** Let  $y_1, y_2, y_3, y_4, y_5, y_6$  denote the dual variables corresponding to the six primal constraints respectively.

$$\min \quad 20y_1 + 35y_2 + 15y_3 + 10y_4 + 5y_5 + 2y_6 \quad (20)$$

$$\text{s.t.} \quad 2y_1 + 7y_2 + 4y_3 + y_4 \geq 2 \quad (21)$$

$$-3y_1 + 2y_2 + 5y_3 + y_5 \geq -7 \quad (22)$$

$$-5y_1 + 6y_2 - 3y_3 - y_6 \geq 6 \quad (23)$$

$$-4y_1 - 2y_2 - 2y_3 \geq 5 \quad (24)$$

$$y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0, y_4, y_5, y_6 \geq 0 \quad (25)$$

## 1.2 (b) KKT Optimality Conditions

### 1.2.1 First Linear Program

$$y_1 * (x_1 + x_2 + x_3 + x_4 - 19) = 0 \quad (26)$$

$$y_2 * (4x_2 + 8x_4 - 55) = 0 \quad (27)$$

$$y_3 * (x_1 + 6x_2 - x_3 - 7) = 0 \quad (28)$$

$$x_1 * (y_1 + y_3 - 5) = 0 \quad (29)$$

$$x_2 * (y_1 + 4y_2 + 6y_3 - 1) = 0 \quad (30)$$

$$x_3 * (y_1 - y_3 + 4) = 0 \quad (31)$$

$$x_4 * (y_1 + 8y_2) = 0 \quad (32)$$

### 1.2.2 Second Linear Program

Complementary Slackness:

$$y_1 * (2x_1 - 3x_2 - 5x_3 - 4x_4 - 20) = 0 \quad (33)$$

$$y_2 * (7x_1 + 2x_2 + 6x_3 - 2x_4 - 35) = 0 \quad (34)$$

$$y_3 * (4x_1 + 5x_2 - 3x_3 - 2x_4 - 15) = 0 \quad (35)$$

$$y_4 * (x_1 - 10) = 0 \quad (36)$$

$$y_5 * (x_2 - 5) = 0 \quad (37)$$

$$y_6 * (x_3 - 2) = 0 \quad (38)$$

$$x_1 * (2y_1 + 7y_2 + 4y_3 + y_4 - 2) = 0 \quad (39)$$

$$x_2 * (-3y_1 + 2y_2 + 5y_3 + y_5 + 7) = 0 \quad (40)$$

$$x_3 * (-5y_1 + 6y_2 - 3y_3 - y_6 - 6) = 0 \quad (41)$$

$$x_4 * (-4y_1 - 2y_2 - 2y_3 - 5) = 0 \quad (42)$$

## 2 Problem 2

### 2.1 a: Linear Programming Formulation

#### 2.1.1 Problem Context

LGUAir must determine optimal seat allocation for three fare classes across three demand scenarios. This is a **two-stage decision problem**: first allocate seats (strategic), then sell tickets based on realized demand (operational).

#### 2.1.2 Decision Variables

- **First-stage variables:**  $x_i$  = number of seats allocated to class  $i$  (fixed across all scenarios)
  - $i = 1$ : first-class
  - $i = 2$ : business-class
  - $i = 3$ : coach-class
- **Second-stage variables:**  $y_{i,j}$  = number of tickets sold for class  $i$  in scenario  $j$ 
  - $j = 1$ : scenario (i) - weekday peak
  - $j = 2$ : scenario (ii) - weekend
  - $j = 3$ : scenario (iii) - weekday off-peak

### 2.1.3 Parameters

- Space usage coefficients:  $a_i$  for class  $i$ , where  $i \in \{1, 2, 3\}$
- Profit coefficients:  $r_i$  for class  $i$ , where  $i \in \{1, 2, 3\}$
- Total capacity:  $C$  (in coach-seat units)
- Demand limits:  $d_{i,j}$  for class  $i$  in scenario  $j$

### 2.1.4 Objective Function

Maximize total profit across all scenarios:

$$\max Z = \sum_{j=1}^3 \sum_{i=1}^3 r_i y_{i,j} \quad (43)$$

### 2.1.5 Constraints

**Space allocation constraint** (first-stage):

$$\sum_{i=1}^3 a_i x_i \leq C \quad (44)$$

**Seat limit constraints** (linking first and second stage):

$$y_{i,j} \leq x_i, \quad \forall i \in \{1, 2, 3\}, j \in \{1, 2, 3\} \quad (45)$$

**Demand constraints** (second-stage):

$$y_{i,j} \leq d_{i,j}, \quad \forall i \in \{1, 2, 3\}, j \in \{1, 2, 3\} \quad (46)$$

**Non-negativity and integer constraints:**

$$x_i \in \mathbb{Z}_{\geq 0}, \quad y_{i,j} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \quad (47)$$

## 2.2 b: Solution Using COPT

The integer linear program was solved using COPT with the Python file “p3\_primal\_corrected.py”.

**Optimal Objective Value**  $Z^* = 625$

**Optimal First-Stage Decision (Seat Allocation)**

**Optimal Second-Stage Decisions (Ticket Sales)** The space constraint is binding (utilization = 100%), indicating optimal capacity usage.

| Class          | Allocated Seats $x_i^*$ | Space Usage  | Total Space |
|----------------|-------------------------|--------------|-------------|
| First-class    | 10                      | 20.0         |             |
| Business-class | 20                      | 30.0         |             |
| Coach-class    | 150                     | 150.0        |             |
| <b>Total</b>   |                         | <b>200.0</b> | <b>200</b>  |

## 2.3 c: Shadow Prices and Dual Formulation

### 2.3.1 Theoretical Note

Since the primal problem has integer constraints, **strong duality does NOT hold**. The dual problem we formulate is the dual of the **LP relaxation** of the original integer program. Therefore, the dual problem's optimal solution provides a **tight upper bound** for the primal ILP optimal value:

$$Z_{\text{ILP}}^* \leq Z_{\text{LP}}^* = W_{\text{Dual}}^* \quad (48)$$

### 2.3.2 Dual Variables

- $\pi \geq 0$ : dual variable for the space allocation constraint
- $\mu_{i,j} \geq 0$ : dual variable for seat limit constraint  $y_{i,j} \leq x_i$
- $\lambda_{i,j} \geq 0$ : dual variable for demand constraint  $y_{i,j} \leq d_{i,j}$

### 2.3.3 Dual Objective Function

$$\min W = C \cdot \pi + \sum_{i=1}^3 \sum_{j=1}^3 d_{i,j} \lambda_{i,j} \quad (49)$$

### 2.3.4 Dual Constraints

For each first-stage variable  $x_i$ :

$$a_i \pi \geq \sum_{j=1}^3 \mu_{i,j}, \quad \forall i \in \{1, 2, 3\} \quad (50)$$

For each second-stage variable  $y_{i,j}$ :

$$\mu_{i,j} + \lambda_{i,j} \geq r_i, \quad \forall i \in \{1, 2, 3\}, j \in \{1, 2, 3\} \quad (51)$$

### 2.3.5 Dual Solution

Solving the dual problem using COPT (file “p3\_dual\_corrected.py”) yields:

**Dual Optimal Value**  $W^* = 625.0$

### Shadow Price (Marginal Value of Capacity)

$$\pi^* = 2.6667 \quad (52)$$

**Economic Interpretation:** The shadow price  $\pi = 2.6667$  represents the marginal value of one additional unit of space capacity. Each unit increase in  $C$  would increase the objective value by approximately  $\pi$  (within the validity range).

**Dual Variables for Seat Limit Constraints** Selected values of  $\mu_{i,j}^*$  (full results in computational output):

- $\mu_{1,1} = 3.0$  (binding),  $\mu_{1,2} = 2.3333$  (binding),  $\mu_{1,3} = 0.0$  (slack)
- $\mu_{2,1} = 2.0$  (binding),  $\mu_{2,2} = 2.0$  (binding),  $\mu_{2,3} = 0.0$  (slack)
- $\mu_{3,1} = 1.0$  (binding),  $\mu_{3,2} = 1.0$  (binding),  $\mu_{3,3} = 0.6667$  (binding)

**Dual Variables for Demand Constraints** Selected values of  $\lambda_{i,j}^*$ :

- First-class:  $\lambda_{1,2} = 0.6667$ ,  $\lambda_{1,3} = 3.0$  (both binding due to tight demand)
- Business-class:  $\lambda_{2,3} = 2.0$  (binding in scenario 3)
- Coach-class:  $\lambda_{3,3} = 0.3333$  (binding in scenario 3)

### Duality Gap

$$\text{Gap} = W_{\text{Dual}}^* - Z_{\text{ILP}}^* = 625.0 - 625.0 = 0.0 \quad (53)$$

In this problem, the LP relaxation solution happens to be integral, so the duality gap is zero and the LP relaxation provides the exact optimal value.

## 2.4 d: Prediction for $C = 201$

Using the shadow price  $\pi = 2.6667$  from part (c), when capacity increases from  $C = 200$  to  $C = 201$ , we predict:

$$\Delta Z = \pi \cdot \Delta C = 2.6667 \times 1 = 2.6667 \quad (54)$$

Therefore, the predicted optimal value is:

$$Z^*(201) \approx Z^*(200) + \Delta Z = 625 + 2.6667 \approx 627.67 \quad (55)$$

**Note:** Since we have integer constraints, the actual increase may differ slightly from this linear approximation.

## 2.5 e: Verification with $C = 201$

Resolving the integer program with  $C = 201$  using COPT (file “p3\_primal\_201.py”) yields:

**Actual Optimal Value**  $Z^*(201) = 627$

### Comparison with Prediction

- Predicted value:  $\approx 627.67$
- Actual value: 627
- Actual increase:  $\Delta Z = 627 - 625 = 2$

**Analysis** The actual increase ( $\Delta Z = 2$ ) is close to but slightly less than the predicted increase (2.6667). This discrepancy occurs because:

1. The shadow price  $\pi$  is derived from the LP relaxation
2. Integer constraints impose additional restrictions
3. The linear approximation is valid only for small changes in  $C$

The prediction provides a good approximation and validates that the shadow price captures the marginal value of capacity within the relevant range.

## 2.6 f: Interpretation of Shadow Prices and Complementary Slackness

### 2.6.1 Complementary Slackness Conditions

For the primal-dual pair, complementary slackness states:

#### Primal Complementary Slackness

$$\mu_{i,j} > 0 \Rightarrow y_{i,j} = x_i \quad (\text{seat limit binding}) \quad (56)$$

$$\lambda_{i,j} > 0 \Rightarrow y_{i,j} = d_{i,j} \quad (\text{demand limit binding}) \quad (57)$$

#### Dual Complementary Slackness

$$x_i > 0 \Rightarrow a_i \pi = \sum_{j=1}^3 \mu_{i,j} \quad (\text{dual constraint tight}) \quad (58)$$

$$y_{i,j} > 0 \Rightarrow \mu_{i,j} + \lambda_{i,j} = r_i \quad (\text{dual constraint tight}) \quad (59)$$

### 2.6.2 Economic Interpretation

**Shadow Price**  $\pi = 2.6667$  The space capacity has high marginal value. Increasing total capacity by 1 coach-seat unit would increase total profit by approximately \$2.67 (averaged across scenarios).

### Seat Limit Dual Variables $\mu_{i,j}$

- When  $\mu_{i,j} > 0$ : The seat allocation constraint is binding in scenario  $j$ . Allocating more seats to class  $i$  would increase profit in that scenario.
- When  $\mu_{i,j} = 0$ : The seat allocation has slack in scenario  $j$ . Allocated seats exceed sales.

### Example: First-Class in Scenario 3

- $\mu_{1,3} = 0$ : Seat limit not binding (allocated 10 seats, sold only 5)
- $\lambda_{1,3} = 3.0$ : Demand limit is binding (demand = 5, sales = 5)
- Interpretation: Increasing seat allocation to first-class would not help in scenario 3 because demand is the bottleneck, not seat availability.

### Demand Dual Variables $\lambda_{i,j}$

- When  $\lambda_{i,j} > 0$ : Demand constraint is binding. Increasing demand would increase profit (if seats are available).
- When  $\lambda_{i,j} = 0$ : Demand constraint has slack. Current seat allocation or other constraints prevent selling more tickets.

### Example: Coach-Class in Scenario 1

- $\lambda_{3,1} = 0$ : Demand constraint not binding (demand = 200, sales = 150)
- $\mu_{3,1} = 1.0$ : Seat limit is binding (allocated = sales = 150)
- Interpretation: Cannot increase sales in scenario 1 even though demand exists, because all allocated coach seats are already sold. The bottleneck is seat allocation, not demand.

## 2.6.3 Managerial Insights

1. **Capacity is valuable:** With  $\pi = 2.6667$ , increasing aircraft capacity would be profitable if the cost is less than \$2.67 per coach-seat unit.
2. **Binding constraints by scenario:**
  - Scenarios 1 & 2: High demand, most seat limits are binding
  - Scenario 3: Lower demand, several constraints have slack
3. **Strategic allocation trade-offs:** The optimal allocation ( $x_1^* = 10, x_2^* = 20, x_3^* = 150$ ) balances:
  - High profit from first-class ( $r_1 = 3$ ) vs. high space usage ( $a_1 = 2$ )
  - Varying demand across scenarios

4. **Revenue management opportunities:** Positive  $\lambda_{i,j}$  values indicate scenarios where demand stimulation (e.g., marketing) could increase profits if seat availability permits.

### 3 Problem 3

#### 3.1 (a) Dual Formulation

**Primal (Maximization):**

$$\max \quad 4x_1 + 2x_2 + 3x_3 \quad (60)$$

$$\text{s.t.} \quad 2x_1 + 3x_2 + x_3 \leq 12 \quad (61)$$

$$x_1 + 4x_2 + 2x_3 \leq 10 \quad (62)$$

$$3x_1 + x_2 + x_3 \leq 10 \quad (63)$$

$$x_1, x_2, x_3 \geq 0 \quad (64)$$

**Dual (Minimization):**

$$\min \quad 12y_1 + 10y_2 + 10y_3 \quad (65)$$

$$\text{s.t.} \quad 2y_1 + y_2 + 3y_3 \geq 4 \quad (66)$$

$$3y_1 + 4y_2 + y_3 \geq 2 \quad (67)$$

$$y_1 + 2y_2 + y_3 \geq 3 \quad (68)$$

$$y_1, y_2, y_3 \geq 0 \quad (69)$$

#### 3.2 (b) Optimal Dual Solution via Complementary Slackness

Given optimal primal solution:  $x^* = (2, 0, 4)$ .

Substituting  $x^*$  into primal constraints:

- Constraint 1:  $2(2) + 3(0) + 1(4) = 8 < 12$  (slack  $\neq 0$ )
- Constraint 2:  $1(2) + 4(0) + 2(4) = 10$  (binding)
- Constraint 3:  $3(2) + 1(0) + 1(4) = 10$  (binding)

By complementary slackness:  $y_1^* = 0$ , and since  $x_1^* = 2 > 0$  and  $x_3^* = 4 > 0$ , dual constraints 1 and 3 must be binding.

With  $y_1 = 0$ :

$$y_2 + 3y_3 = 4 \quad (70)$$

$$2y_2 + y_3 = 3 \quad (71)$$

Solving:  $y_2 = 1, y_3 = 1$ .

Therefore,  $y^* = (0, 1, 1)$ .

### 3.3 (c) Strong Duality Verification

Primal objective value:  $Z_P = 4(2) + 2(0) + 3(4) = 20$

Dual objective value:  $Z_D = 12(0) + 10(1) + 10(1) = 20$

Since  $Z_P = Z_D = 20$ , strong duality holds.

## 4 Problem 4

### 4.1 a: Integer Programming Formulation

Let  $x_{i,j}$  denote a binary decision variable that equals 1 if the drill moves directly from hole  $i$  to hole  $j$ , and 0 otherwise, where  $i, j \in \{1, 2, 3, 4, 5\}$  and  $i \neq j$ . Let  $d_{i,j}$  denote the distance (in millimeters) between hole  $i$  and hole  $j$ . Define auxiliary variables  $u_i$  for  $i = 2, 3, 4, 5$  to eliminate subtours using the MTZ formulation.

**Objective Function** Minimize the total drilling distance:

$$\min Z = \sum_{i=1}^5 \sum_{\substack{j=1 \\ j \neq i}}^5 d_{i,j} x_{i,j} \quad (72)$$

**Constraints**

$$\sum_{\substack{j=1 \\ j \neq i}}^5 x_{i,j} = 1, \quad i = 1, 2, 3, 4, 5 \quad (73)$$

$$\sum_{\substack{i=1 \\ i \neq j}}^5 x_{i,j} = 1, \quad j = 1, 2, 3, 4, 5 \quad (74)$$

$$u_i - u_j + 4x_{i,j} \leq 3, \quad \forall i, j \in \{2, 3, 4, 5\}, i \neq j \quad (75)$$

$$x_{i,j} \in \{0, 1\}, \quad i \neq j \quad (76)$$

$$1 \leq u_i \leq 4, \quad i = 2, 3, 4, 5 \quad (77)$$

### 4.2 b: Solution Using COPT

The integer program was solved using COPT with the Python file "p4.py", and the optimal drilling sequence and distance are as follows:

**Optimal Drilling Sequence (Hole-to-Hole Path)**

$$(1, 5) \rightarrow (5, 4) \rightarrow (4, 2) \rightarrow (2, 3) \rightarrow (3, 1)$$

### Minimum Total Distance

$$Z^* = 35.0 \text{ mm}$$

**Interpretation for Manufacturing Engineer** Start drilling at hole 1, then move to hole 5, continue to hole 4, then to hole 2, next to hole 3, and finally return to hole 1. This sequence minimizes the total travel distance of the drill, achieving the fastest possible drilling process.

## 5 Problem 5

### 5.1 a: Integer Programming Formulation

Let  $y_i \in \{0, 1\}$  denote whether course  $i$  is selected

**Objective Function** Minimize the total number of courses selected:

$$\min Z = \sum_{i=1}^7 y_i \quad (78)$$

**Constraints** Requirement constraints (at least 2 courses for each category):

$$y_1 + y_2 + y_3 + y_4 + y_7 \geq 2 \quad (79)$$

$$y_2 + y_4 + y_5 + y_7 \geq 2 \quad (80)$$

$$y_3 + y_5 + y_6 \geq 2 \quad (81)$$

Prerequisite constraints:

$$y_3 \leq y_6 \quad (82)$$

$$y_5 \leq y_6 \quad (83)$$

$$y_4 \leq y_1 \quad (84)$$

$$y_7 \leq y_4 \quad (85)$$

$$y_i \in \{0, 1\}, \quad \forall i \in \{0, 1, 2, 3, 4, 5, 6\} \quad (86)$$

**Optimal Course Selection** The integer program was solved using COPTPY with the Python file "p5.py" and the optimal objective value is  $Z^* = 4$ , where the courses: Operations Research ( $y_1 = 1$ ), Data Structures ( $y_2 = 1$ ), Computer Simulation ( $y_4 = 1$ ) and Intro to Programming ( $y_5 = 1$ ) are selected